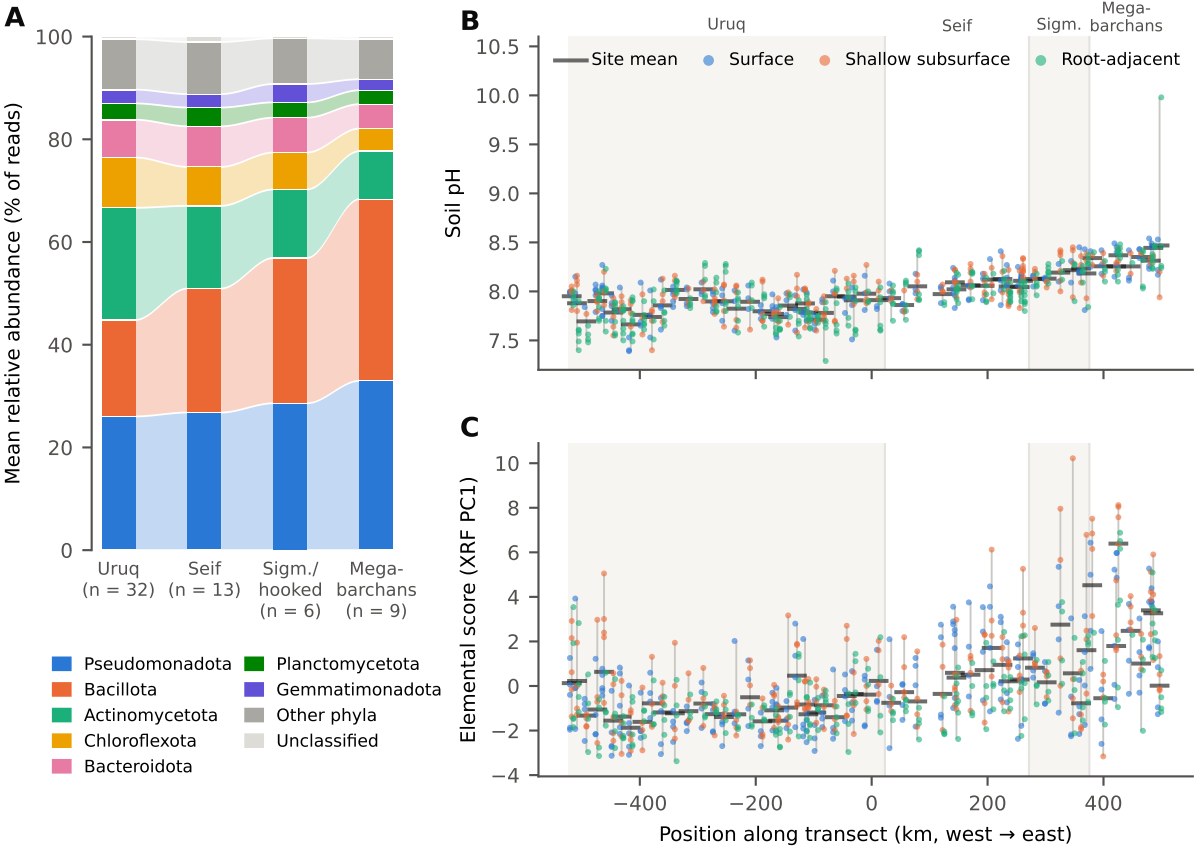

A
Mean relative abundance (% of reads)
100
80
60
40
20
0
Uruq
(n = 32)
Seif
(n = 13)
Sigm./
hooked
(n = 6)
Mega-
barchans
(n = 9)
Pseudomonadota
Bacillota
Actinomycetota
Chloroflexota
Bacteroidota
Planctomycetota
Gemmatimonadota
Other phyla
Unclassified
B
Uruq
Seif
Sigm.
Mega-
barchans
Site mean
Surface
Shallow subsurface
Root-adjacent
Soil pH
10.5
10.0
9.5
9.0
8.5
8.0
7.5
C
Elemental score (XRF PC1)
10
8
6
4
2
0
−2
−4
−400
−200
0
200
400
Position along transect (km, west → east)